

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **C Programming**

Code No: **CSC115** Semester: **1st Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define structure and nested structure. Write a program to find out whether the nth term of the Fibonacci series is a prime number or not. Read the value of n from the user and display the result in the main function. Use separate user-defined function to generate nth Fibonacci term and to check whether that number is prime or not.
2. Explain the relation to array and pointer. Differentiate between call by value and call by reference with a suitable program.
3. Differentiate between source code and object code. Create a structure named Book with members Book_Name, Price and Author_Name, then take input for 10 records of Book and print the name of authors having the price of book greater than 1000.

Section B

Attempt any Eight questions[8×5=40]

4. Describe the different types of I/O functions used in file handling with syntax.
5. Write a program to read P×Q matrix of integers and find the largest integer of each row and display it.
6. Write a program to calculate the factorial of a given number using recursion.
7. Write a program to check whether the entered word is palindrome or not.
8. List different types of operators and explain any three of them.
9. Trace the output

```
#include<stdio.h>
void main(){int i=0, k;for (k=5; k>=0; k--){i = i+k;}printf ("%d\n", i);getch ();}
```
10. Write a program to compute the sum of first 10 even numbers using function.
11. What is dynamic memory allocation? Explain with a suitable program.
12. Write a program to initialize an array of dimension 10 and sent the numbers within the array in ascending order.